11.A Molecular Shape and Bonding Theories I

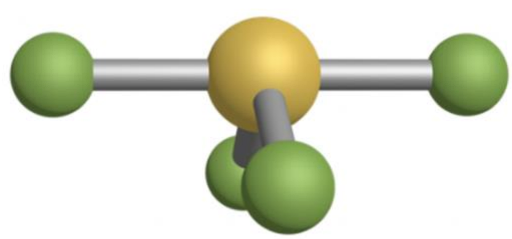
Valence Shell Electron Pair Repulsion (VSEPR) Theory. A great deal of information can be inferred from Lewis structures, if we know what to look for. For example, we can make an educated guess of molecular geometry from a Lewis structure. The principle upon which this determination is based is the simple idea that electron "domains" repel one another, so that domains take up positions that are as far apart as possible.

- How can we predict the three-dimensional geometry of a molecule from its Lewis structure?

- According to **valence shell electron pair repulsion (VSEPR) theory**, regions of electron density **(electron domains)** try to position themselves as far apart from one another as possible to minimize repulsion. Molecular shape follows from this concept.

- Magnitude of repulsion for bonding and nonbonding (lone pair) domains…

11.1. Determine the total number of electron domains, the number of bonding groups, and the number of nonbonding lone pairs for the molecule below. The yellow atom is sulfur and the green atoms are chlorines. To answer this question, enter the number of electron domains at the central atom.

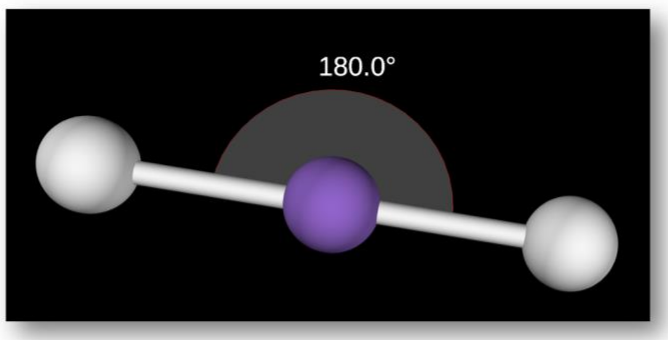


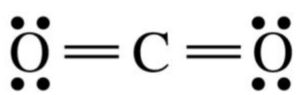
- **Electron geometry** is the arrangement of electron domains around the central atom, irrespective of whether the domains are bonding or nonbonding.

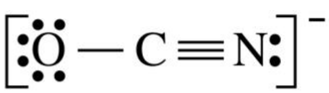
- **Molecular geometry** is the arrangement of *atoms* around the central atom (equivalently, bonding domains only).

- Electron geometry follows from the **steric number** at the central atom. https://phet.colorado.edu/sims/html/molecule-shapes/latest/molecule-shapes_all.html

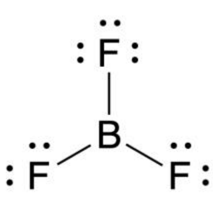
- **Linear** electron geometry (steric number 2)

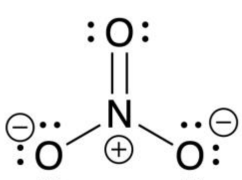


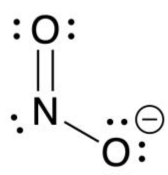


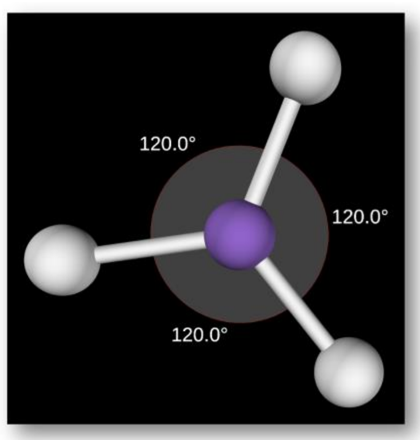


- **Trigonal planar** electron geometry (steric number 3)

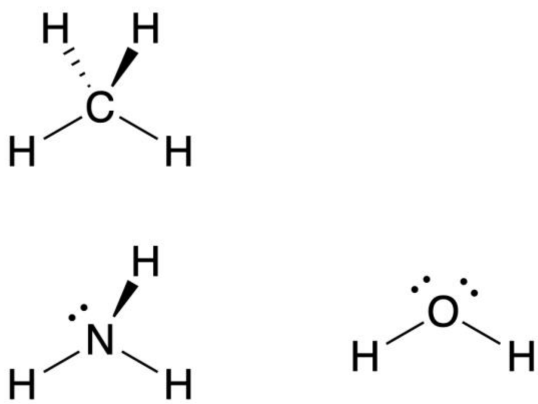


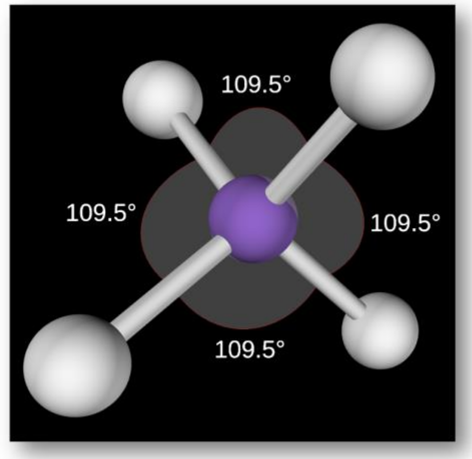




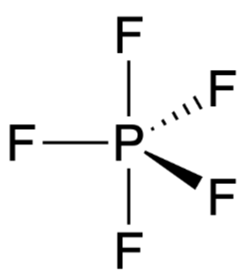


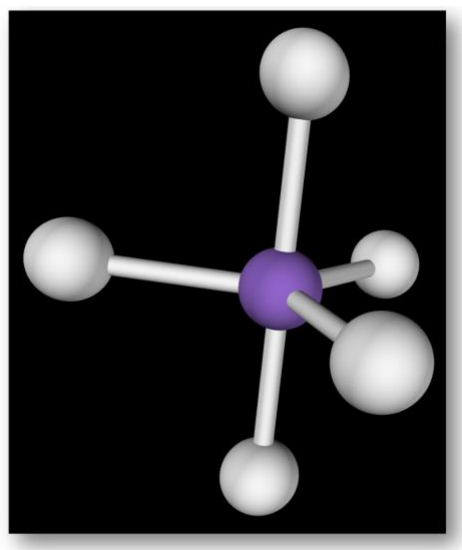
- **Tetrahedral** electron geometry (steric number 4)





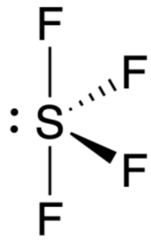
- **Trigonal bipyramidal** electron geometry (steric number 5)
    - Important to distinguish between **axial** and **equatorial** positions

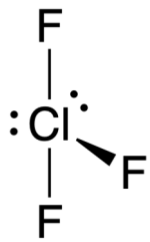


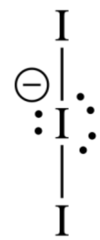


    - For trigonal bipyramidal molecules with lone pairs, where do the lone pairs go? **Example.** Draw a three-dimensional structure for SF_4.

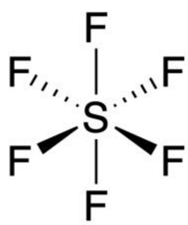
   - Lone pairs occupy the equatorial positions in the most stable structure. This leads to predictable molecular geometries…

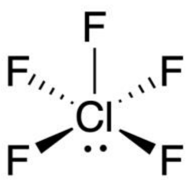


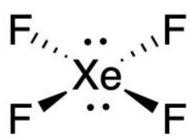


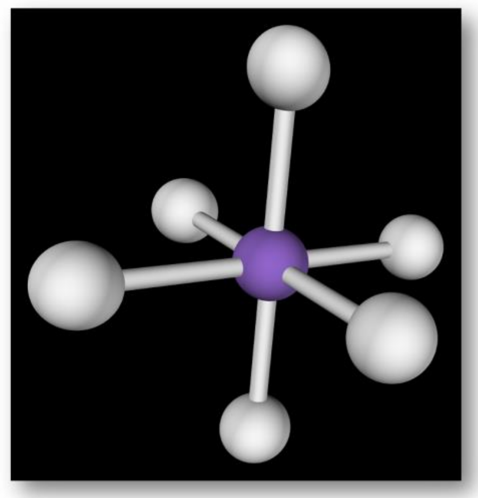


- **Octahedral** electron geometry (steric number 6)



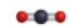
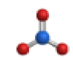
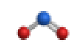
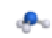
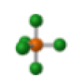
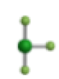
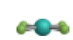




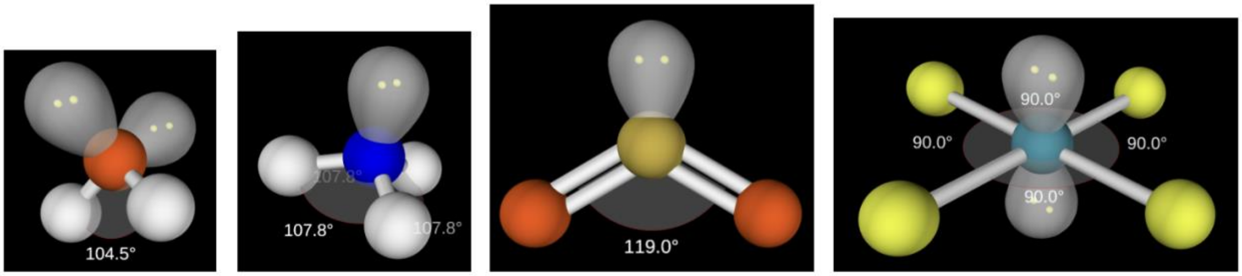


Summary of VSEPR Geometries. Committing to memory the five electron geometries of VSEPR theory and their corresponding steric numbers is a good idea. Molecular geometries can easily be worked out analytically from the electron geometry and number of lone pairs at the atom of interest. The table below summarizes the electron and molecular geometries of VSEPR theory. Note the deviations (“<”) from ideal bond angles—why are these observed?!

TABLE 11.1 Summary of Electron Geometries and Molecular Geometries

Electron Domains	Bonded Groups	Lone Pairs	Electron Geometry	Molecular Geometry	Bond Angles	Example	Model
2	2	0	Linear	Linear	180°	CO_2	
3	3	0	Trigonal planar	Trigonal planar	120°	NO_3^-	
	2	1	Trigonal planar	Bent	< 120°	NO_2^-	
4	4	0	Tetrahedral	Tetrahedral	109.5°	CH_4	
	3	1	Tetrahedral	Trigonal pyramidal	< 109.5°	NH_3	
	2	2	Tetrahedral	Bent	< 109.5°	H_2O	
5	5	0	Trigonal bipyramidal	Trigonal bipyramidal	90°, 120°, 180°	PCl_5	
	4	1	Trigonal bipyramidal	Seesaw	< 90°, < 120°, < 180°	SF_4	
	3	2	Trigonal bipyramidal	T-shaped	< 90°, < 180°	ClF_3	
	2	3	Trigonal bipyramidal	Linear	180°	I_3^-	
6	6	0	Octahedral	Octahedral	90°, 180°	SF_6	
	5	1	Octahedral	Square pyramidal	< 90°, < 180°	ClF_5	
	4	2	Octahedral	Square planar	90°, 180°	XeF_4	

- Lone pairs are larger than bonding pairs. Lone pairs “scrunch” bonding domains closer together, leading to real bond angles slightly less than the VSEPR-ideal angles in molecules containing “asymmetrically positioned” lone pairs.



H_2O NH_3 SO_2 XeF_4

- Note that the effects of symmetrically disposed lone pairs cancel one another!

- For a molecule with multiple central atoms, the overall molecular shape is described as the set of shapes around each atom linked to each other.

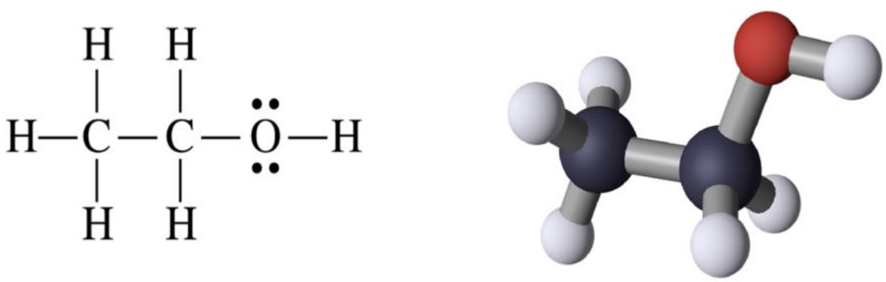


Figure 11.10. Lewis structure and molecular shape of ethanol.

11.2. Select all the molecules from the list below that contain at least one 90° bond angle, considering the *ideal* VSEPR geometries.

A. CH_4

B. PF_5

C. SO_2

D. $SeCl_4$

E. NH_3

11.3. Consider a molecule with bent molecular geometry and a bond angle of 117.5°.

Which statements are *true* regarding this molecule? Select all that apply.

A. The electron geometry is tetrahedral.
B. The electron geometry is trigonal planar.
C. The electron geometry cannot be determined.
D. The molecule has four *total* electron domains.
E. The molecule has three *total* electron domains.

11.B Molecular Shape and Bonding Theories II

11.4. Why does a molecule with 6 regions of electron density and 2 lone pairs adopt a square planar molecular shape? Select all statements that are *true*.

A. The total electron-electron repulsion is minimized by placing lone pairs in equatorial positions.
B. An angle of 180° between lone pairs places them as far away from each other as possible.
C. The bonding domains adopt bond angles of 90° (rather than smaller bond angles) to minimize the electron-electron repulsion amongst themselves.
D. The higher repulsion between bonding domains compared to between lone pairs forces the lone pairs to the axial positions.

Bond Polarity Revisited; Molecular Polarity. In the previous chapter, we noted that in many (most?) covalent bonds, electrons are shared unequally. When electron sharing in a covalent bond is dramatically unequal, the bond is called "polar." However, most bulk properties of compounds relate not to the polarity of individual bonds, but to the polarity—or lack thereof!—of the entire molecule. Here, we'll learn how to combine considerations of bond polarity and geometry to assess whether a molecule is polar or nonpolar. In the next chapter, this determination will inform us on the question of forces *between* molecules.

- When two atoms of different electronegativity engage in a covalent bond, one atom has greater electron density than the other.

- One end of the bond has partial positive charge and the other partial negative charge—a **dipole**.

- **Polar covalent bonds** have substantial dipoles.

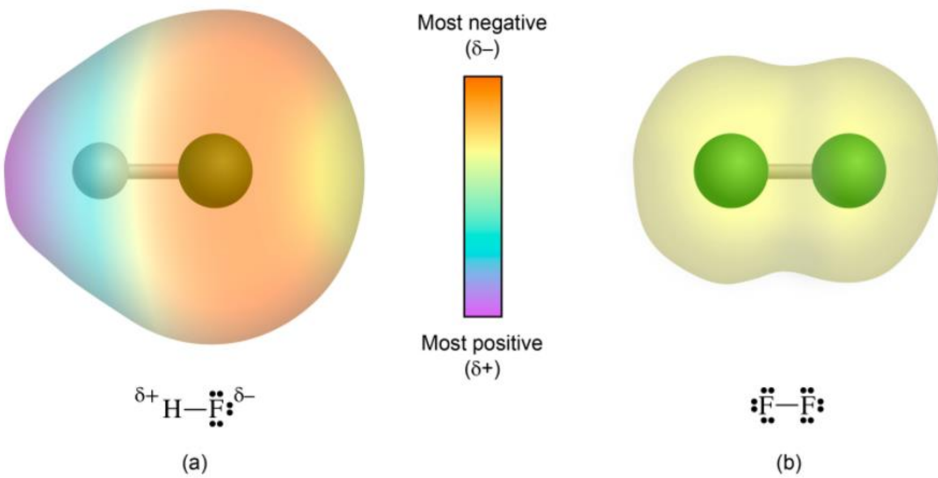


Figure 10.16. (a) Unequal electron sharing in HF and (b) equal electron sharing in F_2.

- **Molecular dipole** can be determined by combining knowledge of…
    -
    -
- Molecules with significant dipole moments are **polar**;
  molecules with zero or negligible dipole moments are **nonpolar**.
- Molecules that contain *only* nonpolar bonds are nonpolar.
- Molecules that contain polar bonds may be polar or nonpolar, depending on geometry!
    - A polar molecule has…
    - A nonpolar molecule has…

- Molecules with two or more polar bonds may be polar or nonpolar, depending on geometry.

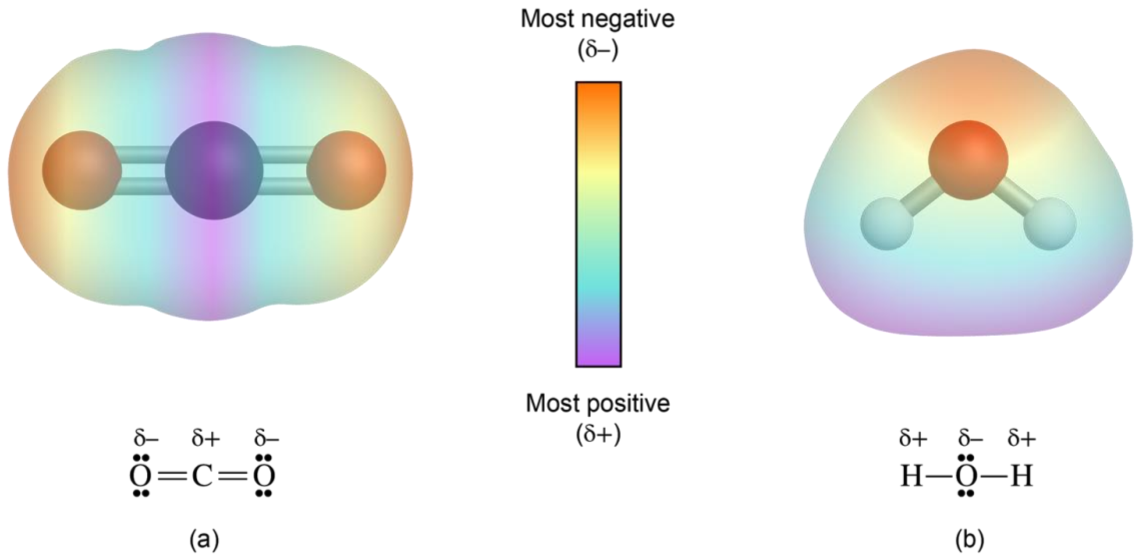


Figure 11.17. Examples of a polar molecule (a) and nonpolar molecule (b) containing polar bonds.

Example. Consider the Lewis structure of CH_2Cl_2. Does this molecule contain polar covalent bonds? Is CH_2Cl_2 polar or nonpolar?

11.5. Consider geometries of the molecules below consistent with VSEPR theory. Which molecules are polar in *all* possible geometries? Enter a string of letters with the necessarily polar molecules in alphabetical order (e.g. “ADEG”).

A. ICl_3

B. CBr_2Cl_2

C. PCl_3F_2

D. XeF_4

E. $XeCl_2F_2$

F. PCl_3

G. ClF_5

11.C Molecular Shape and Bonding Theories III

11.6. Which statement is *true* regarding molecular polarity? For the false statements, generate an example that proves them false.

A. Molecules with lone pairs of electrons on the central atom must be polar.

B. Molecules with square planar molecular geometry must be nonpolar.

C. Tetrahedral molecules with the formula AX_2Y_2 are polar molecules if they contain polar bonds.

D. Molecules for which the electron geometry and molecular geometry are the same must be nonpolar.

E. Molecules that contain only nonpolar bonds can be polar or nonpolar.

Valence Bond Theory. What do electrons "look like" in covalent bonds? Lewis structures provide us with a simple answer, but don't reflect the quantum-mechanical nature of electrons. Bonding theories provide more detailed descriptions of electrons in covalent bonds with more explanatory power, including for example the spatial distributions of bonding electrons and their energies. Valence bond theory is a simple but powerful bonding theory based on the idea that covalent bonds arise from overlap of atomic orbitals.

- Lewis structures convey information about connectivity of atoms.
- VSEPR theory explains the three-dimensional shape of the molecule.
- **Valence bond theory** provides a model for how covalent bonding takes place—specifically how atomic orbitals and electrons are used to construct covalent bonds.
    - Key questions…
        -
        -
        -

- Key aspects of valence bond theory…
  - Orbital overlap
  - Hybrid atomic orbitals

- Covalent bonds are derived from the **overlap** of valence atomic orbitals.

- In diatomic molecules, the atomic orbitals we've already seen overlap to form covalent bonds.

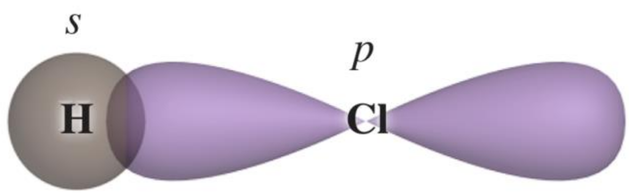


Figure 11.20. Overlap of a 1*s* and 3*p* orbital provides a model of the covalent bond in H—Cl.

- The simple atomic orbitals don't work as well to describe bonding in larger molecules.

- For example, in H_2O we can envision two half-filled 1*s* orbitals on hydrogen overlapping with two half-filled 2*p* orbitals on oxygen.

- What's wrong with this picture?

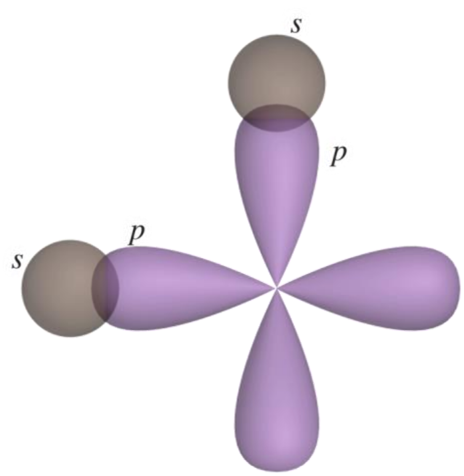


Figure 11.21. Overlap of two 1*s* orbitals with two *p* orbitals suggests two bonds at a 90° angle.

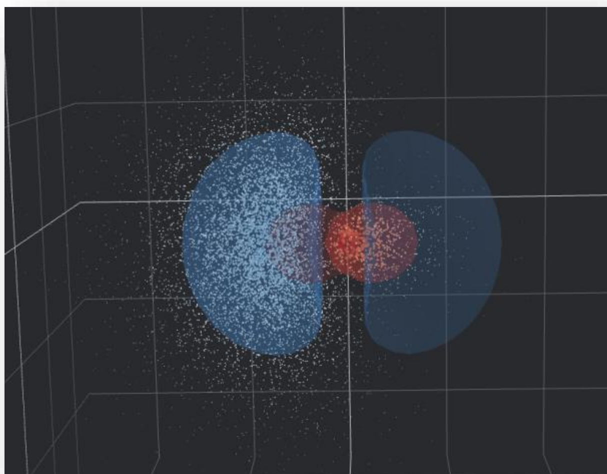
- **Hybrid atomic orbitals** are linear combinations of the atomic orbitals on one atom oriented in directions that are consistent with the VSEPR geometries.

- The number of atomic orbital "inputs" is equal to the number of hybrid orbital "outputs."

- Written as sp^n, where 1 *s* orbital is combined with *n p* orbitals

- Used for sigma bonding and nonbonding lone pairs

https://al2me6.github.io/evanescence/



- ***sp*³ hybridization.** Atoms with four electron domains require four hybrid orbitals, which are constructed from four atomic orbitals (1 *s* + 3 *p*).

    - **Example.** Methane (CH_4)

- To form covalent bonds, the larger lobe of each hybrid orbital overlaps with another hybrid orbital or atomic orbital.

- Nonbonding lone pairs can occupy hybrid orbitals as well.

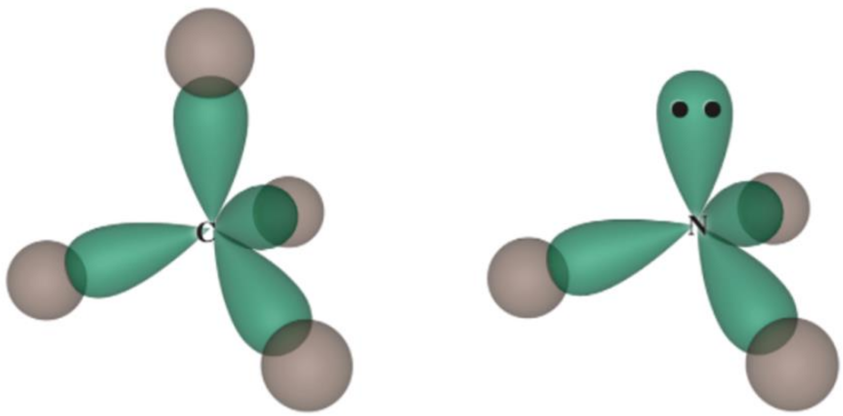


Figure 11.22. Hybrid orbitals (green) overlapping with 1*s* orbitals on hydrogen in CH_4; a lone pair of electrons occupying a hybrid orbital in NH_3.

- ***sp*² hybridization.** Atoms with three electron domains require three hybrid orbitals, which are constructed from three atomic orbitals (1 *s* + 2 *p*).

    - **Example.** Borane (BH_3)

- ***sp* hybridization.** Atoms with two electron domains require two hybrid orbitals, which are constructed from two atomic orbitals (1 *s* + 1 *p*).

    - **Example.** Hydrogen cyanide (HCN)

- Bonds constructed from the "head-on" overlap of two hybrid orbitals or atomic orbitals are called **sigma bonds**.

- All covalent bonds involve one and only one sigma bond.

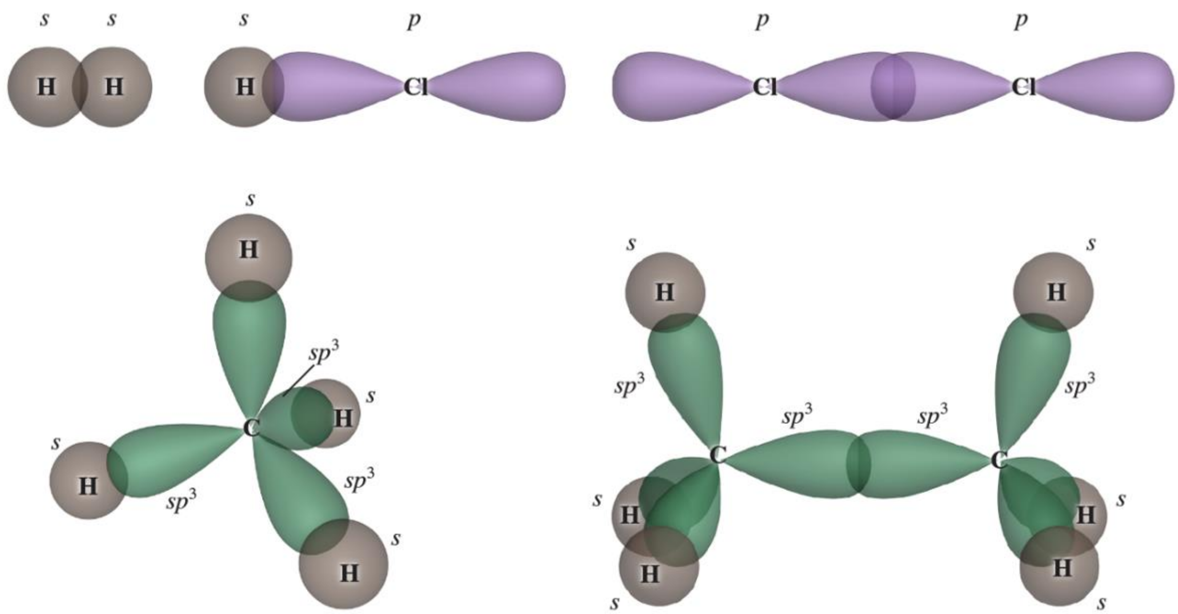
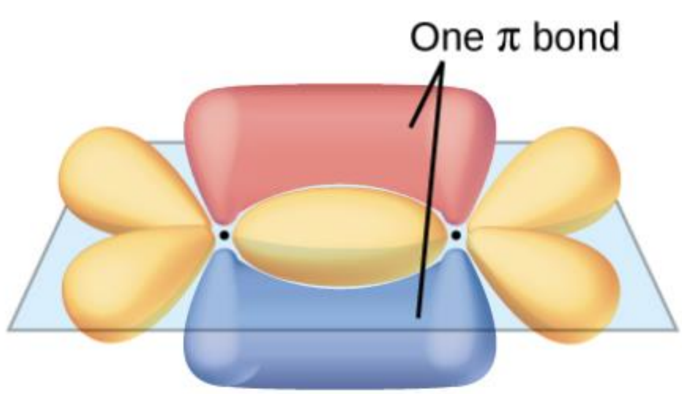


Figure 11.25. Examples of sigma bonds.

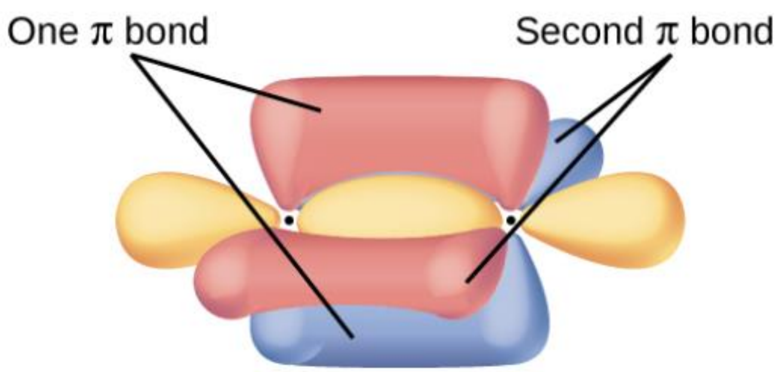
- Unhybridized *p* orbitals (not used to construct hybrids) overlap in a "side-on" manner to form **pi bonds**.

- Pi bonds are the "second and third bonds" of double and triple bonds.



OpenStax Chemistry 2e, Figure 8.23. Sigma bonds (orange) and pi bond (red and blue) in the ethylene molecule (C_2H_4).

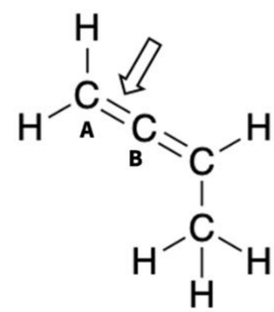
- Two sets of unhybridized *p* orbitals can overlap to form two distinct pi bonds (in a triple bond).



OpenStax Chemistry 2e, Figure 8.25. Sigma bonds (orange) and pi bonds (red and blue) in the ethyne molecule (C_2H_2).

11.7. Determine the hybridizations of the carbon atoms involved in the indicated bond. Complete the following sentence and enter the numeral *xy* as your response.

Carbon **A** has sp^x hybridization and carbon **B** has sp^y hybridization.

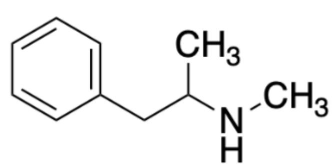


What orbitals overlap to form the sigma bond at the indicated position?

What orbitals overlap to form the pi bond at the indicated position?

11.8. In skeletal structures of organic compounds, carbons are located at the intersections of bonds and C—H bonds are omitted under the assumption that neutral carbons satisfy the octet rule. Formal charges are never omitted.

Add missing C—H bonds to the structure of methamphetamine below and determine the hybridization of each atom. How many sp^3-hybridized atoms does this molecule contain?



- Hybridization does not adequately describe bonding at atoms with more than four electron domains. **Hypervalency** is a more accurate model than sp^3d^n hybridization, but we will not cover it in CHEM 1211K.

- Valence bond theory can be used to explain certain observations of chemical structure and reactivity…

    1. Electrons in double and triple bonds are more reactive than electrons in single bonds.

    2. Single bonds rotate freely, but rotation about double bonds is extremely slow.

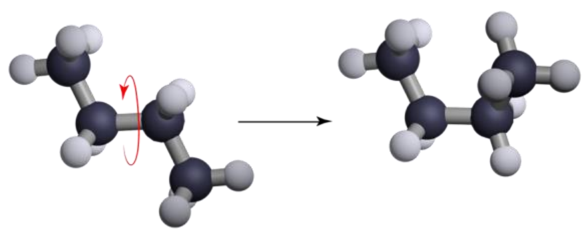


Figure 11.32. The central single bond in butane rotates very rapidly.

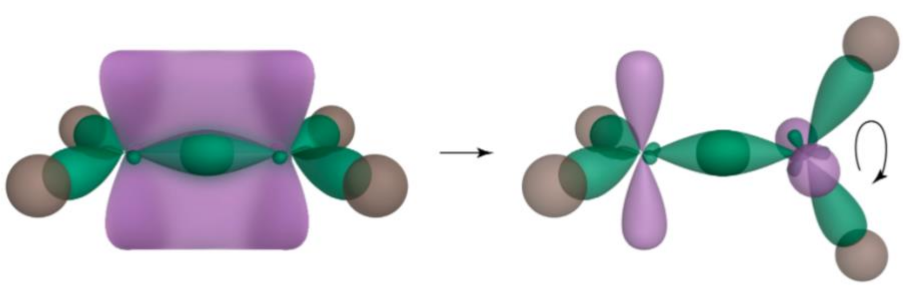


Figure 11.33. Rotation about the double bond in ethylene breaks the pi bond by eliminating orbital overlap.

11.D Molecular Shape and Bonding Theories IV

Limitations of Valence Bond Theory; Introducing Molecular Orbitals. The explanatory power of valence bond theory is limited, especially for molecules exhibiting profound electron delocalization (resonance!) and molecules that display magnetism. For these cases, molecular orbital theory is a more powerful theory. The essential idea of molecular orbital theory is that the Schrödinger equation applies just as well to molecules as to atoms, and that linear combinations of atomic orbitals provide good approximate solutions to molecular Schrödinger equations.

- Valence bond theory does not account for several important observations of molecules, such as…
    - Energy levels of electrons in molecules
    - Magnetic properties
- **Molecular orbital theory** is a bonding theory that explains these properties (and so much more!).
    - Recall that the atomic orbitals $\psi(n, \ell, m_\ell, m_s)$ are solutions to the Schrödinger equation for the hydrogen atom. Are there analogous versions of the Schrödinger equation for molecules? Yes!

$$\hat{H}\psi = E\psi$$

- Your textbook provides a solid overview of molecular orbital theory and additional information can be found in chapter 8 of OpenStax Chemistry 2e.
- Several images in the following lecture slides are from this resource:

  Flowers, P., Theopold, K., Langley, R., & Robinson, W. (2019). Advanced Theories of Covalent Bonding. In *Chemistry 2e*. OpenStax: Houston, TX. Retrieved from https://openstax.org/books/chemistry-2e/pages/preface.

- Atomic orbitals on different atoms overlap to form **molecular orbitals** that are **delocalized** over the entire molecule.

- Important concepts related to molecular orbitals (MOs)…

    - MOs are constructed from AOs on every atom in the molecule (LCAO-MO). For a molecule with *N* valence atomic orbitals on all atoms,[1]

$$\psi_{\mathrm{MO},j} = \sum_{i}^{N} c_i \cdot \psi_{\mathrm{AO},i}$$

    - Because all atoms contribute to all MOs in general, MOs are delocalized.

    - Each MO can only hold two electrons—the Pauli principle still applies. All molecules have more than one MO.

- Atomic orbitals may overlap in an additive way (**constructive**) or subtractive way (**destructive**), analogous to the way two-dimensional waves may interfere constructively or destructively.

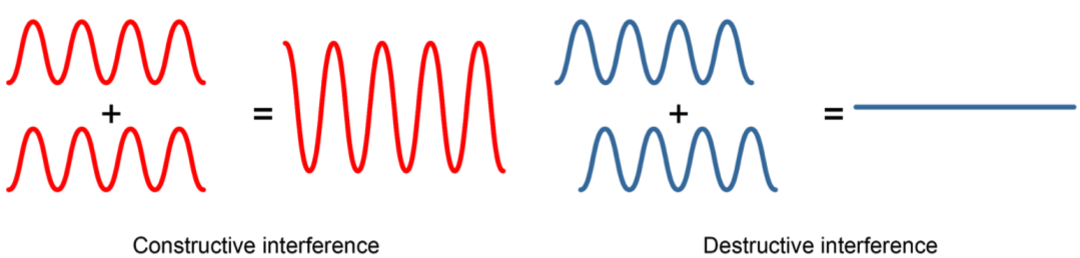


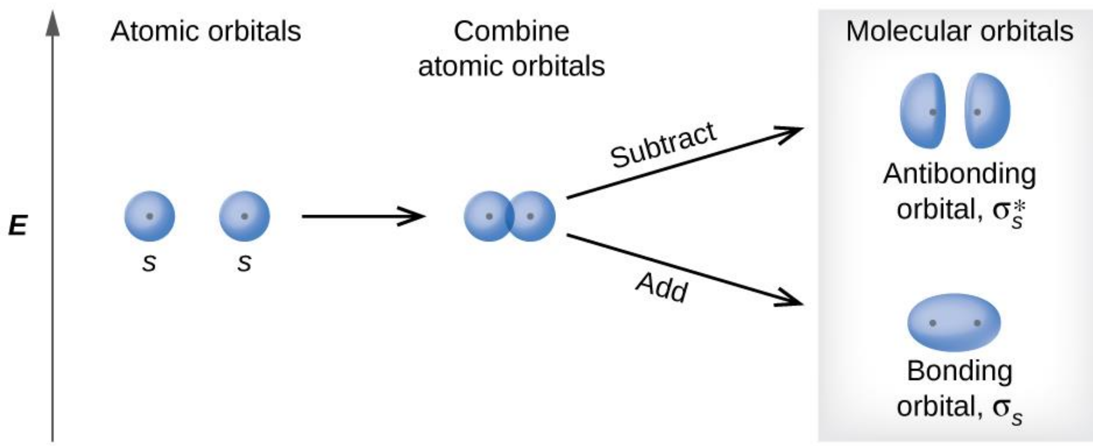
Figure 11.34. Constructive and destructive interference of waves.

[1] This equation is important to understand *conceptually*. The coefficients *c* provide a measure of the "contribution" or "size" of the lobe in the molecular orbital in the vicinity of the atom to which each atomic orbital "belongs." We can see the relative sizes of the coefficients in the shapes of molecular orbitals. However, computers work out the full set of *c*'s that best solve the Schrödinger equation for a given molecule.

Molecular Orbitals of Dihydrogen (H_2). The dihydrogen molecule is a good place to begin with molecular orbital theory, since each hydrogen has only one valence atomic orbital (1*s*). The paradigms developed here for bonding and antibonding MOs will appear repeatedly as we study the molecular orbitals of diatomic molecules.

- Subtractive overlap (destructive interference) of atomic orbitals results in an **antibonding molecular orbital** that is *higher* in energy than the "input" AOs.

- Additive overlap (constructive interference) of atomic orbitals results in a **bonding molecular orbital** that is *lower* in energy than the "input" AOs.



OpenStax Chemistry 2e, Figure 8.29. Combination (overlap) of 1*s* orbitals on hydrogen results in antibonding and bonding molecular orbitals.

- **Molecular orbital (MO) diagrams** show how atomic orbitals (AOs) are combined to form MOs as well as the relative energies of the AOs and MOs and their occupancies.

- MOs formed from "head-on" overlap of AOs are called **sigma orbitals**.

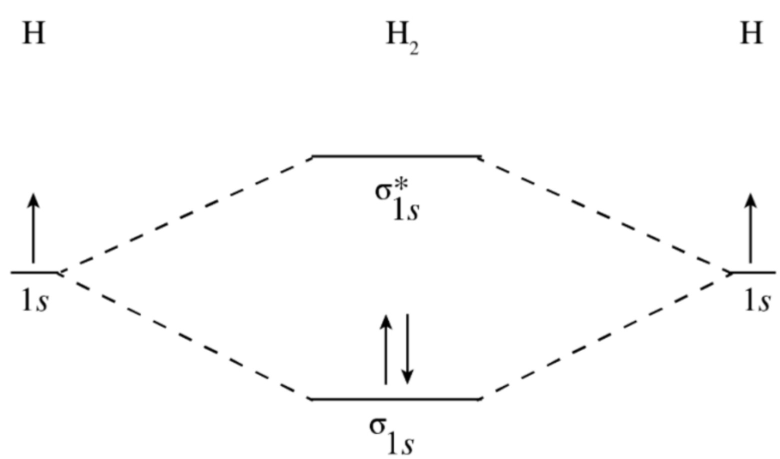


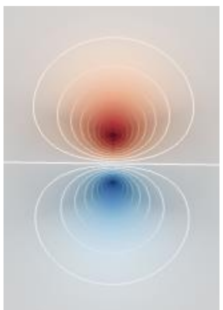
Figure 11.36. Molecular orbital diagram for dihydrogen (H_2).

- We can write molecular electron configurations by analogy to atomic electron configurations…these provide the “address” of each molecular electron.

11.10. Two-dimensional cross-sections of the molecular orbitals of H_2 are shown below. Select all the *true* statements from the choices below.

Hint: nuclei are located at the “densest” points in these orbital shapes.
Color indicates the sign of the wave function.

Orbital 1



Orbital 2

A. Orbital 1 is a σ orbital and orbital 2 is a σ* orbital.

B. Both orbitals are derived from the overlap of 2*p* AOs on the hydrogen atoms.

C. In the neutral H_2 molecule, orbital 1 is empty and orbital 2 contains two electrons.

D. Orbital 2 is higher in energy than the AOs of which it is composed.

E. Orbital 2 contains a node between the nuclei.

- **Bond order** is a measure of the “number of bonds” between two centers.

- For a diatomic molecule, bond order is equal to half the difference between the numbers of bonding and antibonding electrons.

$$\text{bond order} = \frac{\text{bonding } e^- - \text{antibonding } e^-}{2}$$

- Diatomic molecules with a bond order of zero are unstable relative to the separated atoms (cf. He_2).

11.11. Draw a molecular orbital diagram for He_2^+. What is the bond order of this molecule? Enter your response as a numeral.

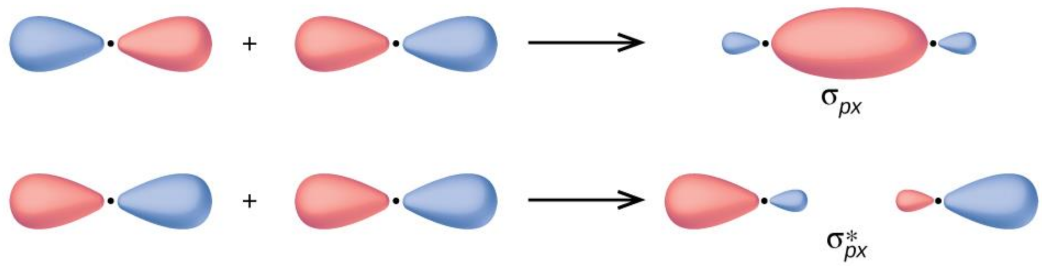
Molecular Orbitals of Homonuclear Diatomic Molecules in the Second Period. When we move from the first period to the second, the situation becomes more complicated because the valence shells of the bonded atoms include 2*s* and 2*p* atomic orbitals. Multiple bonds start appearing in Lewis structures (consider N_2) and we "observe" these in molecular orbital diagrams as pi-type bonding and antibonding molecular orbitals.

- So far we have focused on 1*s* orbitals and the period 1 elements. When the period 2 elements form diatomic molecules, the 2*s* and 2*p* AOs combine to form eight **valence MOs**.

- The 2*p* orbitals overlap to form one pair of sigma orbitals (σ_{2px}, $\sigma^{\star}_{2px}$) and two pairs of **pi orbitals** (π_{2py}, $\pi^{\star}_{2py}$ and π_{2pz}, $\pi^{\star}_{2pz}$).

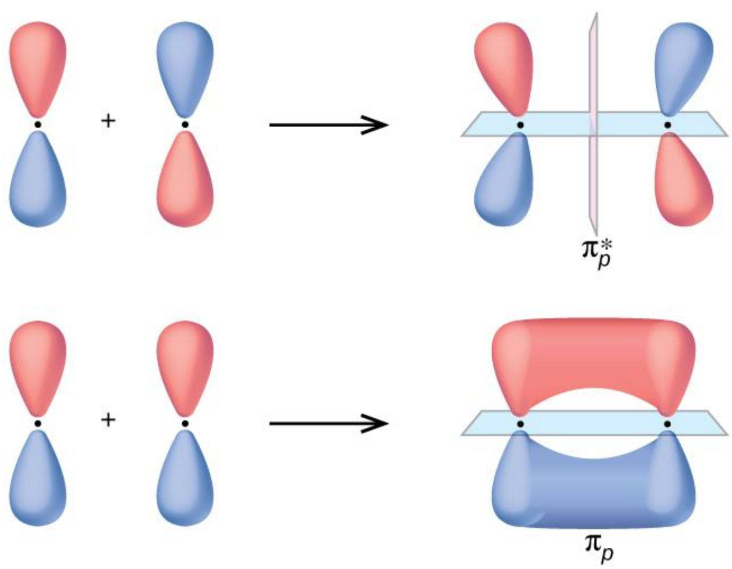
- The 2*s* orbitals overlap to form a pair of sigma orbitals (σ_{2s}, $\sigma^{\star}_{2s}$).

- 2*p* orbitals that overlap "head on" along the internuclear axis form one pair of sigma orbitals (σ_{2px}, σ^*_{2px}).



OpenStax Chemistry 2e, Figure 8.30. Head-on overlap of 2p_x orbitals gives rise to sigma bonding and antibonding MOs.

- 2*p* orbitals that overlap "side on" form two pairs of pi orbitals (π_{2py}, π^*_{2py} and π_{2pz}, π^*_{2pz}).



OpenStax Chemistry 2e, Figure 8.31. Side-on overlap of 2*p* orbitals gives rise to pi bonding and antibonding MOs.

11.E Molecular Shapes and Bonding Theories V

Homonuclear Diatomic Molecules in the Second Period, Continued. We saw above that in diatomic molecules of the second-period elements, such as N_2 and O_2, valence 2*s* and 2*p* atomic orbitals overlap to form molecular orbitals. Now, we'll draw energy diagrams to describe those MOs and see that the energetics of the MOs depends on the electronegativity of the bonded atoms.

- Molecular orbital diagrams for diatomic molecules of the period 2 elements (B_2 through Ne_2)

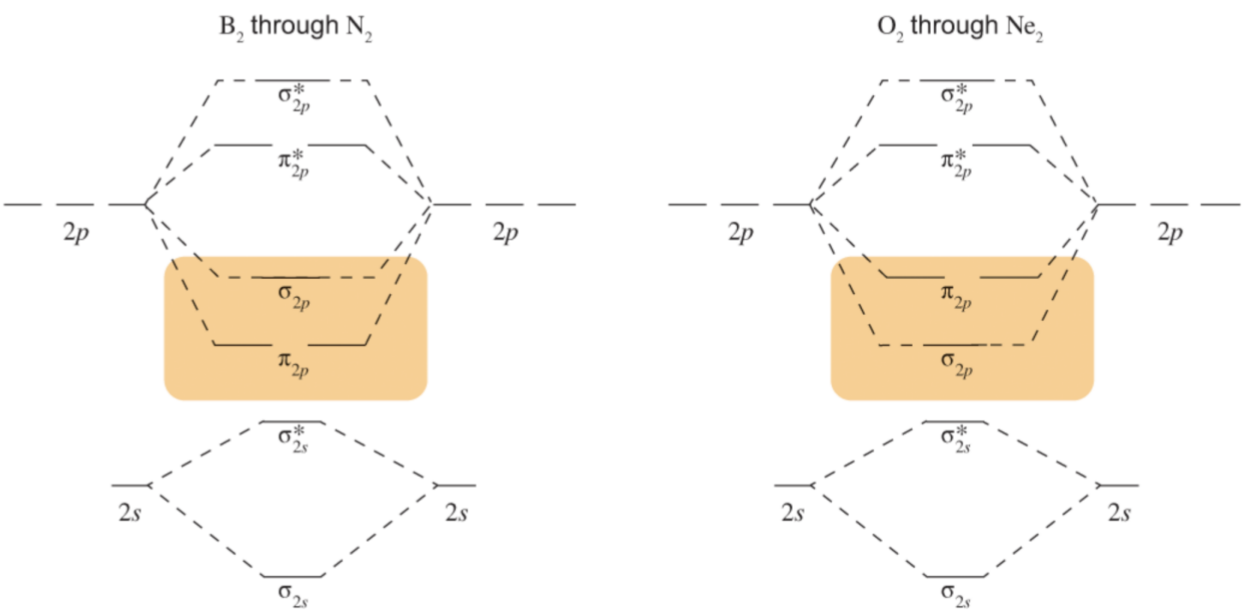
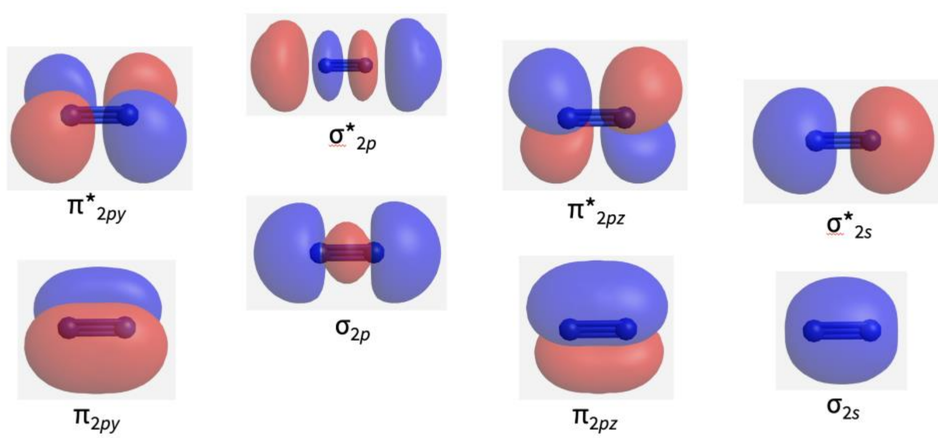
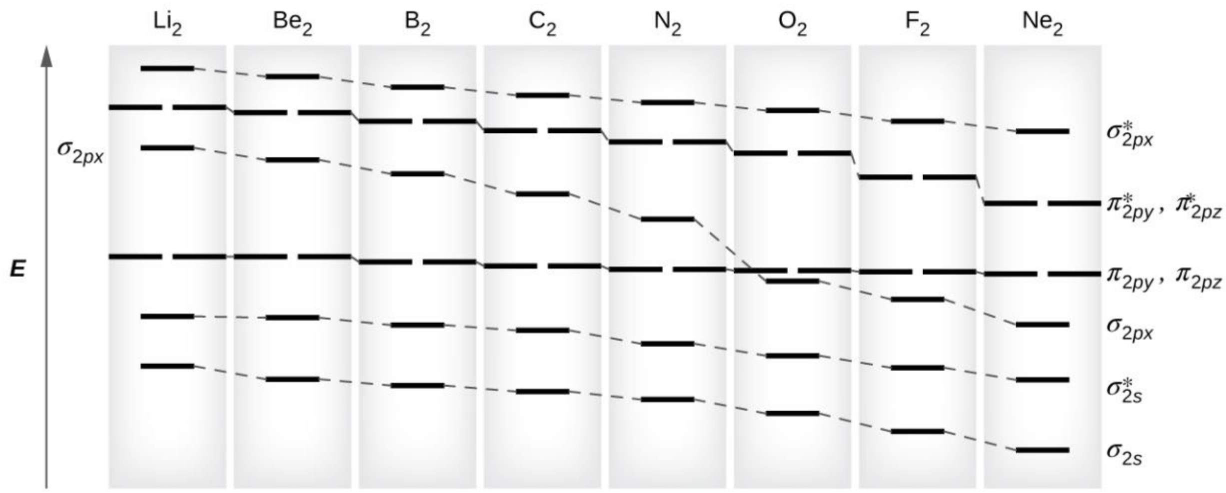


Figure 11.41.[2] Haim, A. *J. Chem. Educ.* **1991**, *68*, 737. (link)



[2] Note that, technically, the lefthand diagram contains an error. The "weird" energetic ordering in B_2 through N_2 is due to *s*-p_x mixing, which affects the energies of the σ bonding orbitals *only*. The σ_{2p} orbital goes up in energy substantially, but the π orbitals are not lowered in energy. The figure on the next page shows the idea properly.

- Periodic trends account for the observed "inversion" in the π_{2p} and σ_{2p} energies…



OpenStax Chemistry 2e, Figure 8.37. Relative energies of orbitals in the second-row diatomic molecules.

- Atoms of relatively *low* electronegativity are affected by s-p_x mixing, which pushes up the energy of the σ_{2px} MO.

- The energetic ordering observed for O_2, F_2, and Ne_2 is "standard."
  $\sigma \rightarrow \pi \rightarrow \pi^* \rightarrow \sigma^*$ is the ordering that will become default in organic chemistry.

Example. Determine the MO electron configuration and bond order of N_2.
Does the bond order agree with the classical Lewis structure?

Example. Determine the MO electron configuration and bond order of O_2.
Does the bond order agree with the classical Lewis structure?

- Removing or adding electrons (oxidation or reduction) can affect bond order.
    - Bond order increases when bonding electrons are added or antibonding electrons removed.
    - Bond order decreases when antibonding electrons are added or bonding electrons removed.
- Excitation of a molecule by light also causes a change in bond order when the occupancies of bonding and antibonding orbitals change.

Example. Which molecule contains a stronger bond, O_2 or O_2^{2+}?
Which molecule contains a longer bond?

11.12. How many electrons does N_2^{2+} have in valence antibonding orbitals?

11.13. Which molecule or ion could have the ground state molecular orbital electron configuration below? Assume that orbitals are listed from lowest to highest energy.

$$(\sigma_{2s})^2(\sigma_{2s}^*)^2(\sigma_{2p})^2(\pi)^4(\pi^*)^3$$

A. F_2
B. N_2^{3-}
C. F_2^+
D. N_2
E. O_2^{2-}

11.F Molecular Shape and Bonding Theories VI

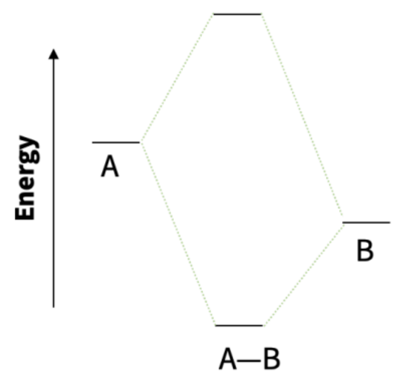
Molecular Orbitals of Heteronuclear Diatomic Molecules. *Heteronuclear* diatomic molecules contain two different atoms. Differences in the energies of overlapping atomic orbitals cause the resulting molecular orbitals to appear "skewed." However, the sense of skew tends to align with our expectations based on electronegativity and bond polarity: filled MOs tend to have larger lobes on more electronegative atoms, while empty orbitals have larger lobes on less electronegative atoms. Molecular orbitals resemble in shape the atomic orbital to which they are closest in energy.

- In a **heteronuclear** diatomic molecule, the energies of overlapping AOs are unequal. MO splitting is relatively small and each MO has greater contribution from one atom or the other.

- The closer in energy an MO is to a contributing AO, the greater the contribution of that AO to the MO.

- Orbitals overlap in a similar way to the homonuclear diatomics but orbital shapes are "skewed."

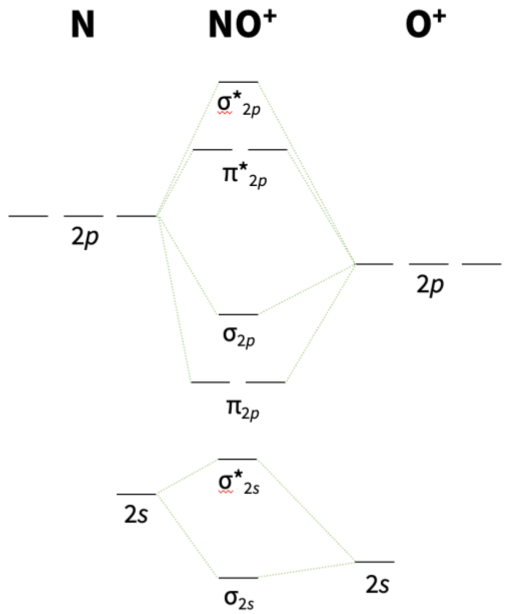
- The MOs are filled according to the Aufbau principle and Hund's rule, as usual for the ground state!



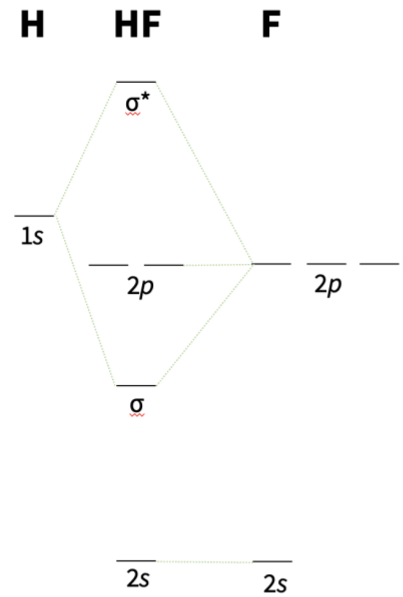
- *s*-*p* mixing and the relative energies of MOs can get tricky with the second-row diatomics; MO diagrams will usually be provided where needed (your job is to *interpret*, not generate).

- H—X diatomics feature purely nonbonding orbitals due to the symmetry of the 1*s* AO (see the next page).

Nitrosonium cation (NO^+)



Hydrogen Fluoride (HF)

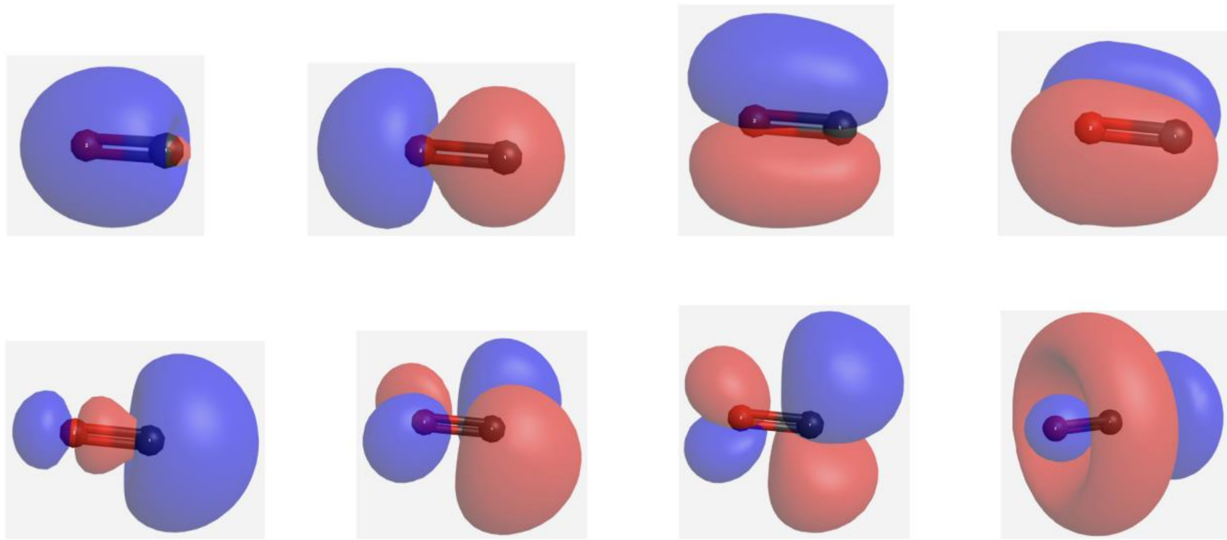


- Why are there three nonbonding orbitals in hydrogen fluoride?

11.14. Draw a molecular orbital diagram for the hydroxide anion (OH^-). Label each orbital as sigma or pi and bonding, antibonding, or nonbonding. For nonbonding orbitals, use the appropriate atomic orbital label.

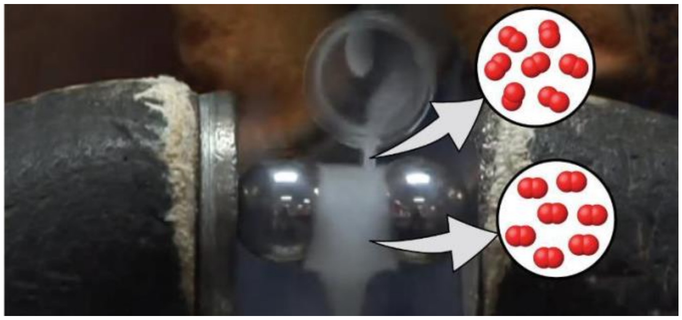
Draw and label the shapes of the MOs. Be sure to account for the “skew” in the orbital shapes based on the difference in electronegativity between oxygen and hydrogen.

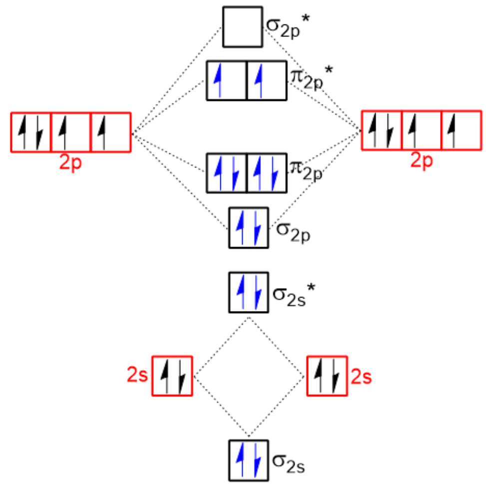
11.15. The molecular orbitals of CO are shown below from lowest to highest energy. Label each orbital as σ or π and bonding or antibonding. Atom 1 is carbon and atom 2 is oxygen. How many pi-type orbitals are shown?



Magnetic Properties. Magnetism in materials is due to the presence of many unpaired electrons. Molecular orbital theory helps us spot unpaired electrons in a way that valence bond theory cannot: by adding electrons to an MO energy diagram using the Aufbau principle and Hund's rule, we can recognize when unpaired electrons are expected in a molecule and predict when attraction to a magnetic field is expected (or not).

- Molecules that contain unpaired electrons exhibit **paramagnetism**, strong attraction to a magnetic field.
- Molecules containing only paired electrons are called **diamagnetic** and display very, very weak repulsion from a magnetic field (basically no useful magnetic properties).
- The MO diagram and molecular electron configuration tell us whether a molecule contains unpaired electrons!





OpenStax Chemistry 2e. Liquid O_2 is attracted to a magnetic field. The molecular orbital energy diagram for O_2 shows that it is expected to contain two unpaired electrons in π* orbitals.

Determine whether each of the molecules below is paramagnetic or diamagnetic.
Consider the molecular electron configuration in each case to justify your conclusion. Select the *paramagnetic* molecules.

A. F_2

B. N_2^{3-}

C. F_2^{+}

D. N_2

E. O_2^{2-}

F. C_2^{+}

Theory	Summary	Uses	Drawbacks
Lewis Theory	Atoms in molecules and polyatomic ions have completed valence shells following the octet/duet rule	Molecular diagrams show relative bond strengths as single, double, and triple bonds.	Two-dimensional model of three-dimensional molecules; no distinction between bonds in different compounds. Does not explain how/why bonds form.
VSEPR	The number of electron domains around the central atom in Lewis structure determines the three-dimensional shape of the molecule or polyatomic ion.	Predicts shapes and can apply polar bonds to determine the polarity of the molecule.	Does not explain how/why bonds form.
Valence Bond Theory	Covalent bonds form by overlap of atomic orbitals in sigma and pi bonds. The degree of overlap determines the strength of the bond.	Description of pi bonds helps to explain greater reactivity of compounds with double bonds and lack of rotation around double bonds. Hybrid atomic orbitals explain some molecular geometries, but not all.	Does not explain certain properties such as paramagnetism. Not easily used to predict properties of hypothetical molecules or ions.
Molecular Orbital Theory	Atomic orbitals combine to form bonding and antibonding molecular orbitals that extend over the entire molecule.	Explains the electron energy levels of molecules, explains paramagnetism/diamagnetism, and makes it possible to predict the properties of hypothetical molecules and ions.	Easily used for diatomic molecules of the first- and second-period atoms; becomes more complex as the size of the molecule increases.

Chapter Summary. Some of the amazing things we can now do with molecular shape and bonding theories…

1. Draw and describe the electron geometries and molecular geometries of valence shell electron pair repulsion (VSEPR) theory.
2. Determine the electron geometry and molecular geometry at an atom that is part of a given or previously deduced Lewis structure.
3. Recognize and explain deviations from “ideal” VSEPR bond angles due to nonbonding lone pairs.
4. Apply valence bond theory to describe covalent bonds as derived from the overlap of atomic orbitals on adjacent atoms.
5. Determine the hybridization of an atom in a given or previously deduced Lewis structure; describe sigma bonds as derived from the overlap of hybrid atomic orbitals.
6. Apply the general principles of molecular orbital theory; recognize a given molecular orbital shape as bonding/antibonding and sigma/pi.
7. Determine the molecular electron configuration for a given homonuclear diatomic molecule or ion; determine bond order from molecular electron configuration.
8. Interpret the molecular orbitals of heteronuclear diatomic molecules or ions.
9. Infer whether a given diatomic molecule or ion is paramagnetic or diamagnetic.